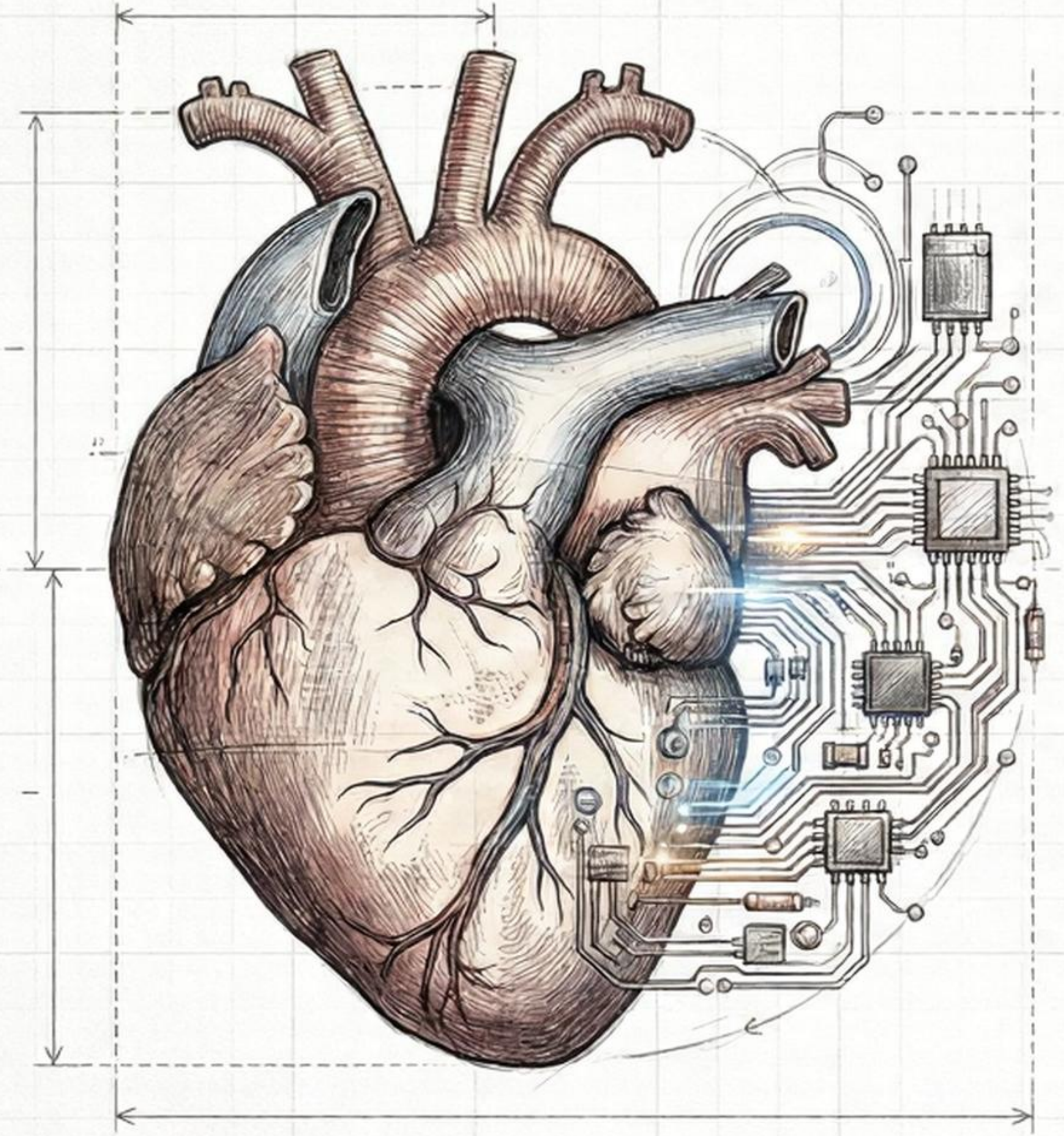


Predictive Intelligence in Healthcare

End-to-End Hospital Readmission Prediction using Databricks Lakehouse

Introducing a vision where data science and machine learning are integrated into medical decision-making to improve patient outcomes. This presentation showcases a specific application of predictive intelligence, focusing on a robust architecture for a critical industry challenge.

Engineered by:

ILAKKIYAN A V

Role: Analytics Engineer

Connect:
https://www.linkedin.com/in/ilakkiyan-av/

The Paradigm Shift: Descriptive vs. Predictive

Moving Beyond "What Happened?"

Descriptive (The Old Way)

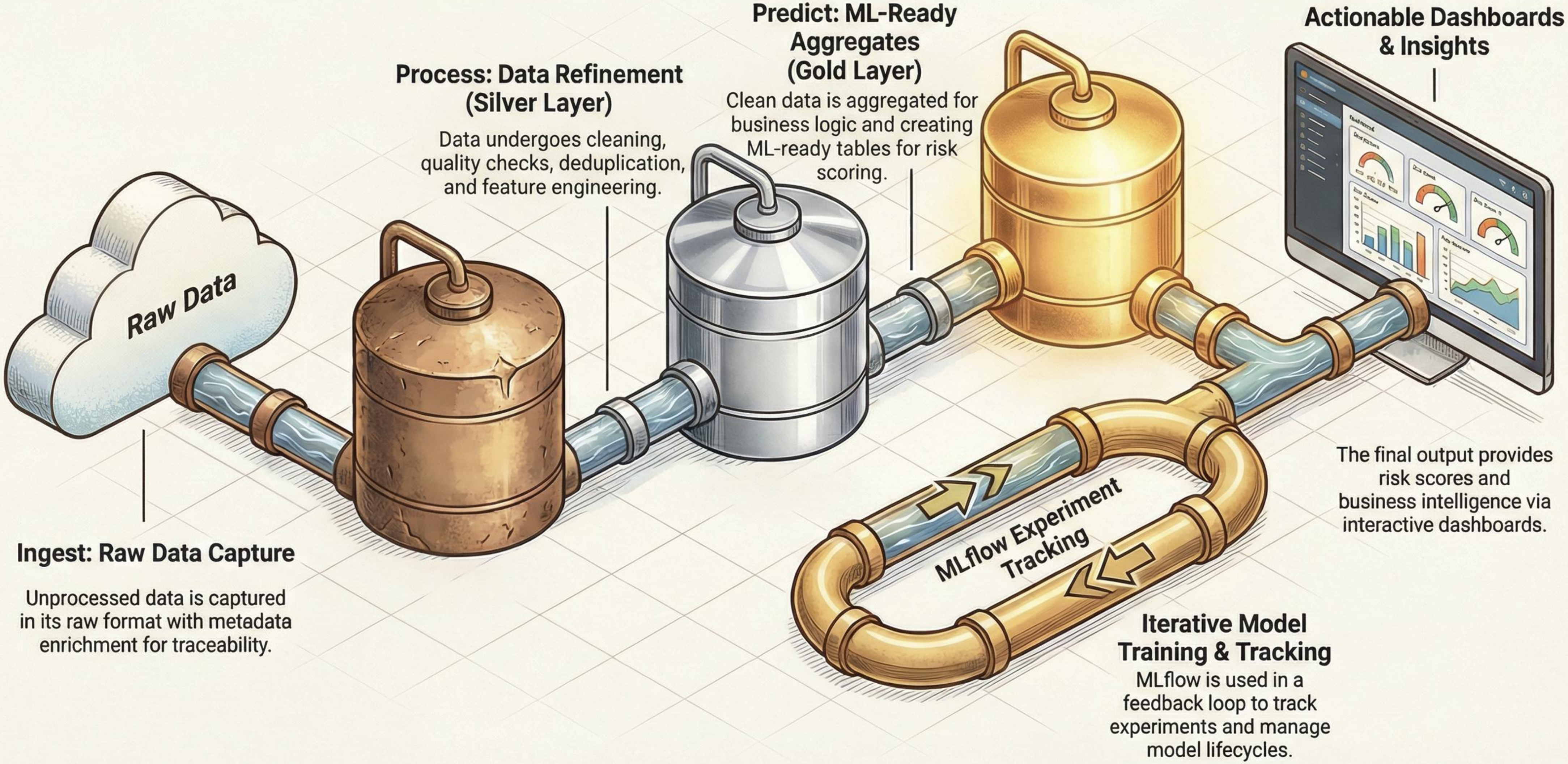
Queries answer "How many readmissions occurred?" (**Reactive**).

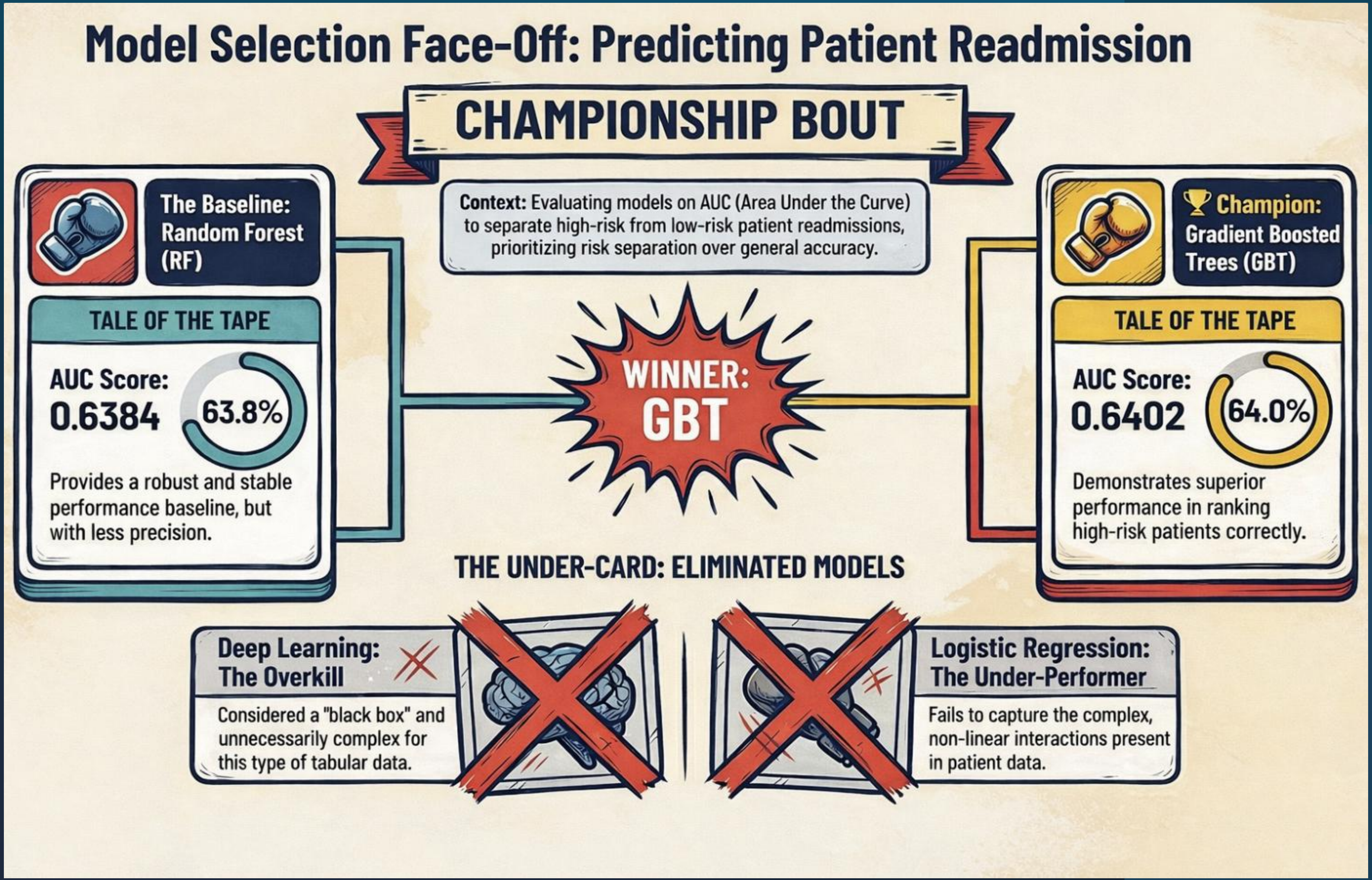
Predictive (The New Way)

Models answer "Who is at risk next?" (**Proactive**).

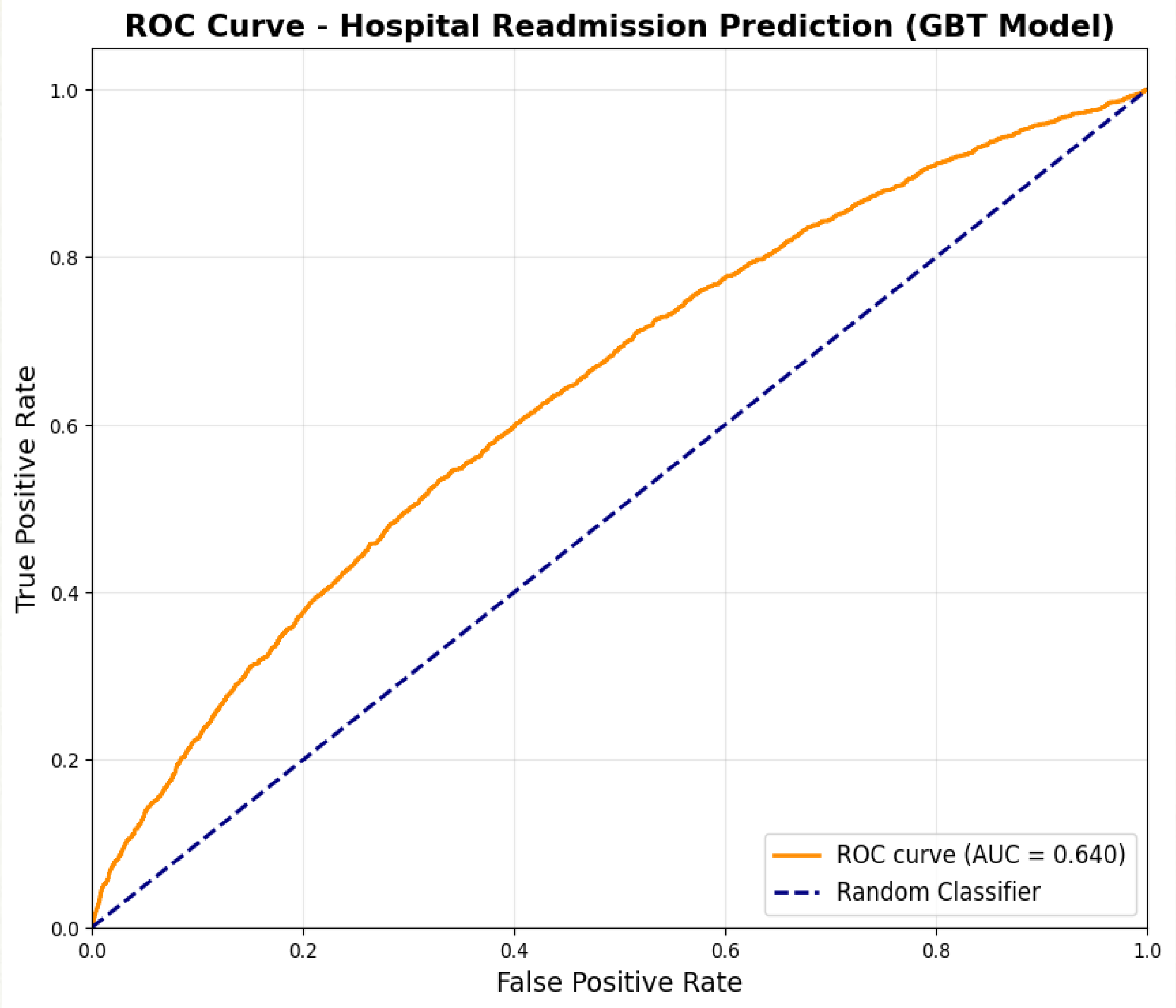
The Goal: Early intervention and efficient resource allocation.

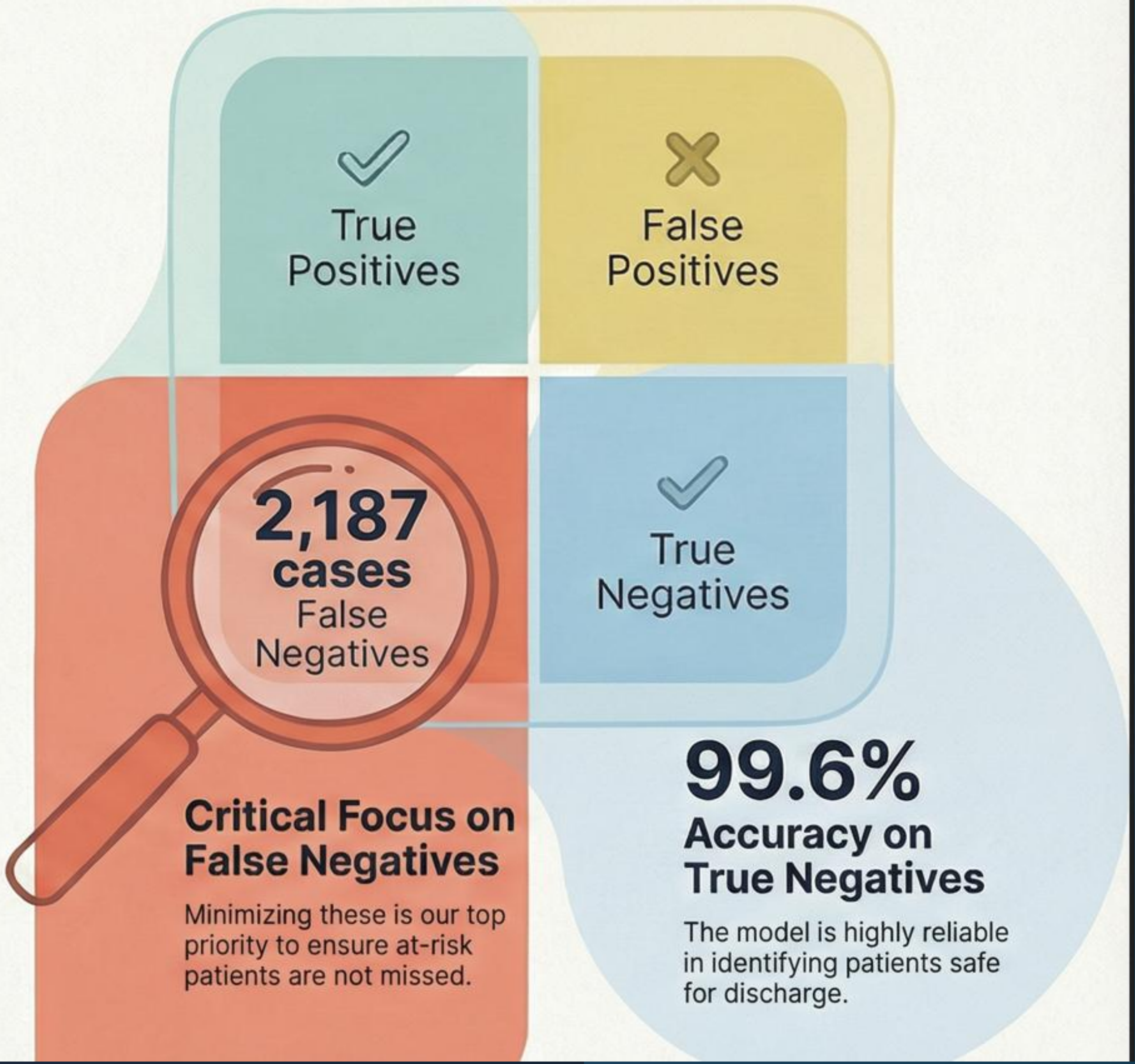
A Unified Data Journey: The Lakehouse Architecture




Model Selection Face-Off: Predicting Patient Readmission
CHAMPIONSHIP BOUT
The Baseline: Random Forest (RF)
TALE OF THE TAPE
AUC Score: 0.6384
63.8%
Provides a robust and stable performance baseline, but with less precision.
Context: Evaluating models on AUC (Area Under the Curve) to separate high-risk from low-risk patient readmissions, prioritizing risk separation over general accuracy.
WINNER: GBT
Champion: Gradient Boosted Trees (GBT)
TALE OF THE TAPE
AUC Score: 0.6402
64.0%
Demonstrates superior performance in ranking high-risk patients correctly.
THE UNDER-CARD: ELIMINATED MODELS
Deep Learning: The Overkill
Considered a "black box" and unnecessarily complex for this type of tabular data.
Logistic Regression: The Under-Performer
Fails to capture the complex, non-linear interactions present in patient data.

Model Performance Validation: Predicting Patient Readmission





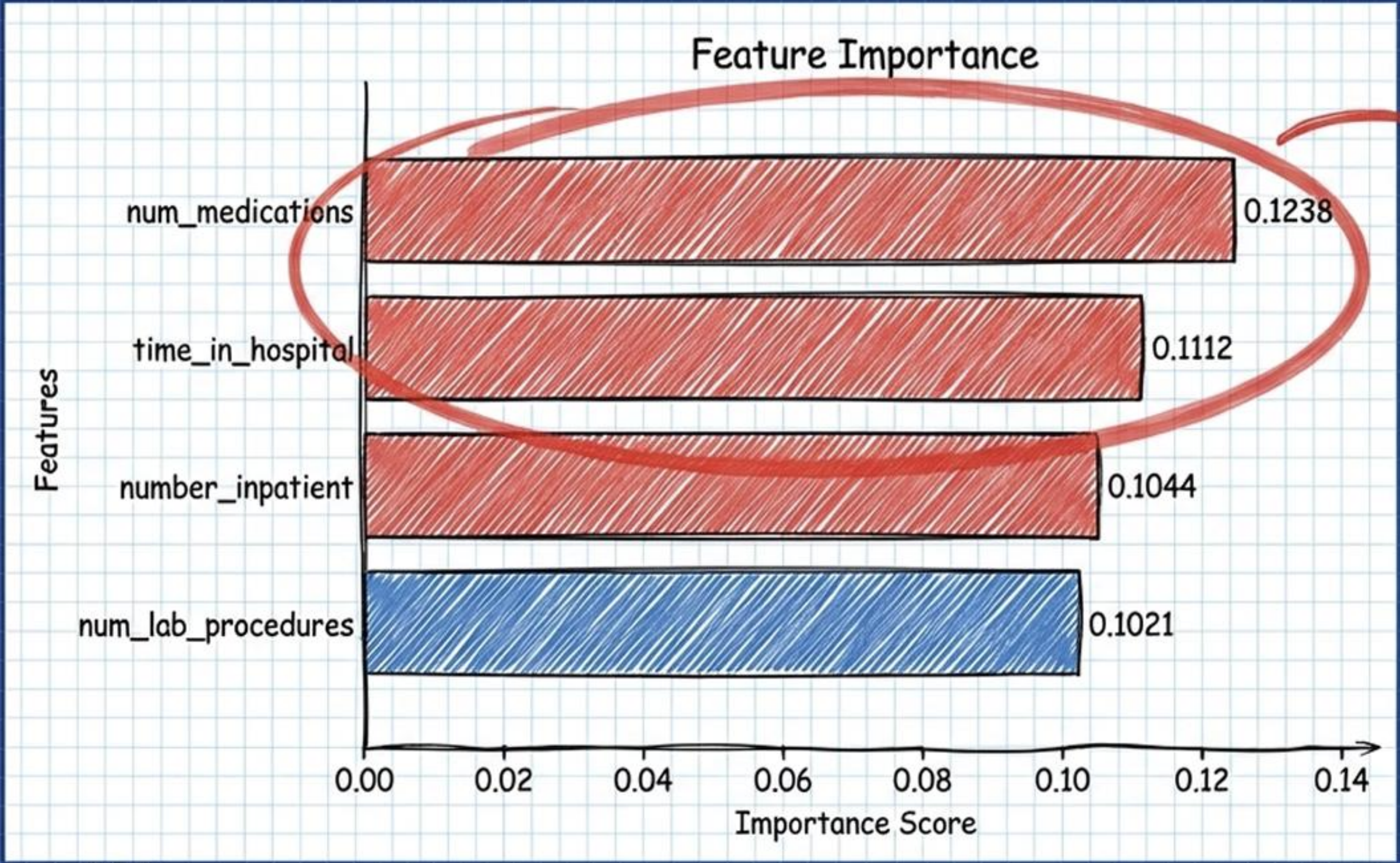
Alternative Model Analysis

Why We Ruled Out Other Architectures

Model	Why Not Chosen
~~Logistic Regression~~	Assumes linearity; misses complex interaction effects.
~~Deep Learning~~	Overkill for tabular data; low interpretability (**Black Box**).
~~SVM~~	Hard to scale; difficult to tune for probability outputs.
~~KNN~~	Sensitive to noise; computationally **slow** at inference.
~~Naive Bayes~~	Unrealistic independence assumptions for medical history.

“ ” **Final Takeaway:** Tree-based ensemble models provide the best balance of performance, robustness, and explainability for structured healthcare data.

DECODING RISK: EXPLAINABILITY



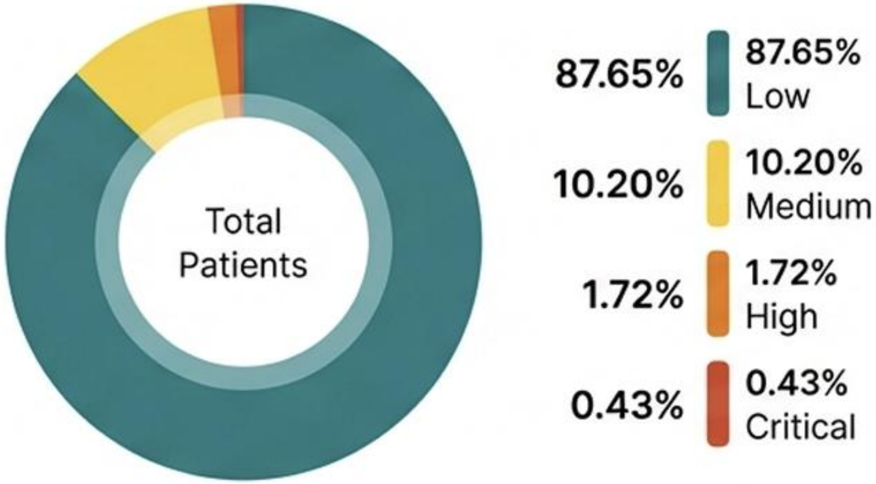
It's not just age.
It's the **INTENSITY** of treatment that drives risk.

Operational Intelligence: The Risk Dashboard

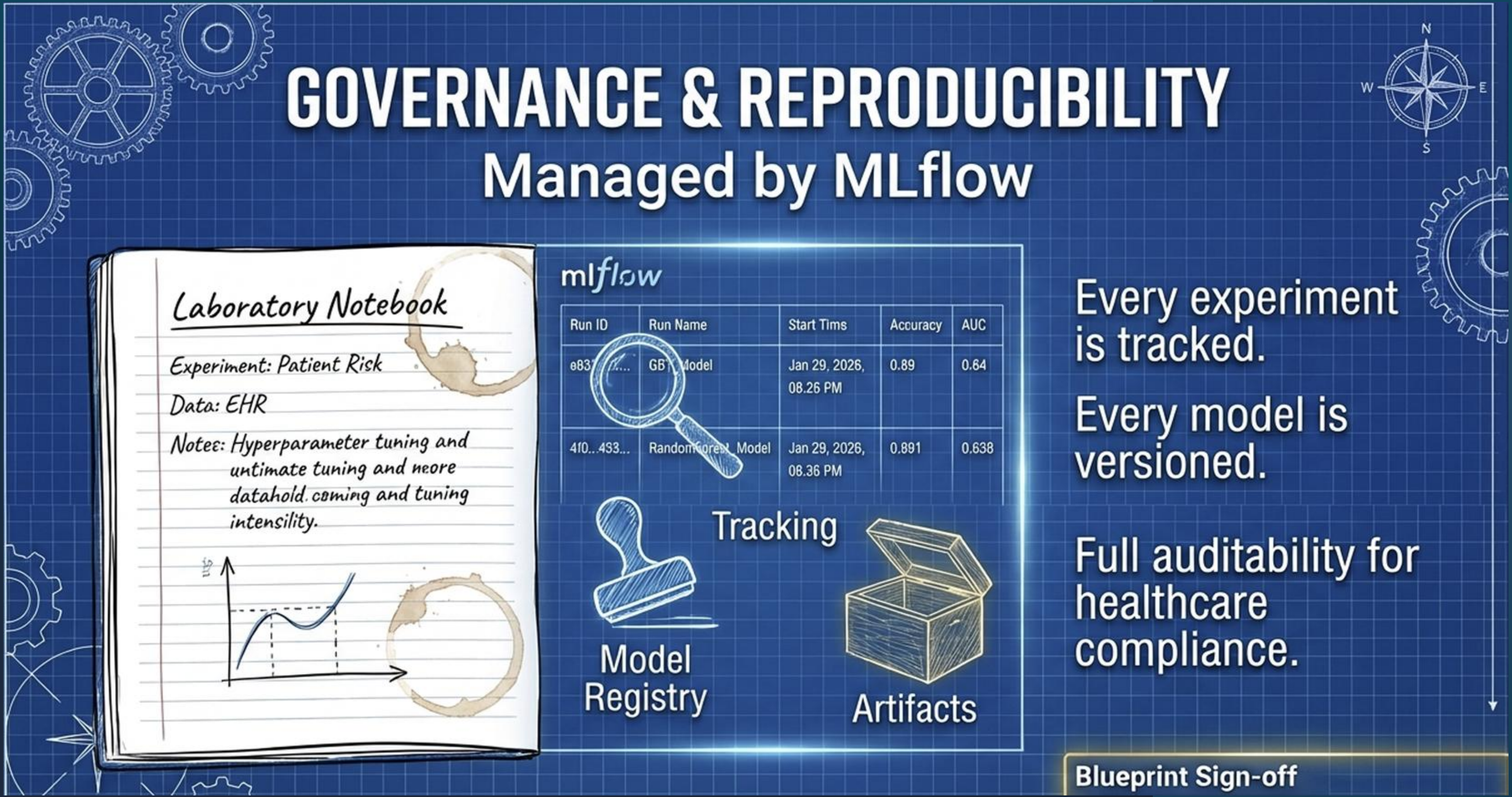
Turning Scores into Decisions

Filter by: Age Group | Department   Surgical Teal

Risk Tier



Patient ID	Risk Tier	Recommendation
P-12345	Low	Routine Follow-up
P-67890	Medium	Telehealth Check-in
P-11223	High	Care Manager Review
P-44556	Critical	Urgent Intervention
P-77889	Low	Routine Follow-up


GOVERNANCE & REPRODUCIBILITY
Managed by MLflow
Laboratory Notebook
Experiment: Patient Risk
Data: EHR
Notes: Hyperparameter tuning and untimate tuning and neore datahold coming and tuning intensility.
mlflow
Run ID
Run Name
Start Tims
Accuracy
AUC
Jan 29, 2026, 08.26 PM
0.89
0.64
4f0...433...
Jan 29, 2026, 08.36 PM
0.891
0.638
Tracking
Model Registry
Artifacts
Every experiment is tracked.
Every model is versioned.
Full auditability for healthcare compliance.
Blueprint Sign-off

Conclusion & Future Impact

A Foundation for AI-Driven Healthcare

Scalable

Architecture handles growing data volumes.

Explainable

Clinicians understand why a patient is flagged.

Actionable

Direct integration into daily workflows.

“GBT was selected for superior risk discrimination, while Random Forest served as a strong baseline.”

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